1. Introduction

The Biden administration's 2022 CHIPS Act explicitly subsidizes semiconductor investment in allied countries, assuming political alignment attracts technology Foreign Direct Investment(FDI). This paper tests whether that assumption holds empirically. The underlying premise is clear: being more politically aligned with a country attracts FDI, especially in the technology sector where security is a huge concern. Regardless of this policy shift, the evidence for the friendshoring hypothesis is limited.

This paper focuses on a specific question: Does geopolitical alignment with the United States influence where American firms invest? The paper examines whether political alignment with the United States in the United Nations General Assembly leads to an outsized share of the U.S. outflow of foreign direct investments to the country's Science and Technology sector. Using an unbalanced panel of 58 countries observed annually from 1999 to 2024, and industry-level FDI data from the U.S. Bureau of Economic Analysis (BEA, 2024) scaled by host country GDP to ensure cross-country comparability, I find a robust negative association between UN ideal point distance and U.S. Science and Technology FDI as a share of host country GDP, meaning that countries more closely aligned with the United States receive systematically more technology investment. This result holds under country fixed effects and survives the inclusion of year fixed effects, providing stronger identification than is typical in this literature. Importantly, a pooled OLS estimate yields a positive and significant coefficient, a sign reversal driven by the strong correlation between alignment and GDP per capita that OLS cannot fully absorb which underscores why within-country variation is the appropriate basis for inference. No comparable alignment effect is detected in Finance or Mining under equivalent specifications, suggesting that the relationship is specific to the Science and Technology sector rather than a general feature of U.S. outward FDI. These findings are best understood as progress-stage results that establish a credible, sector-specific baseline association between geopolitical alignment and technology investment, and motivate a more complete empirical strategy focused on causal identification.

This question matters economically for two reasons. Firstly, Science and Technology FDI is different from other forms of investment. It is not just money or resources that move but productive capacities, implicit knowledge, and innovative capabilities that have a significant impact on growth prospects and structural changes over time. Depending on who receives this flow of productive capacities, different economies will develop their industrial base differently. Therefore, the process becomes political in nature since geopolitical factors determine who will receive the most advanced productive capacities, resulting in different rates of economic development between nations. Secondly, the recent U.S. policy on foreign investment has explicitly recognized this connection. For example, the scope of CFIUS review expanded under the Foreign Investment Risk Review Modernization Act (FIRRMA,2018), targeting those areas where the country saw threats to its technological superiority, namely science and technology. Recently, the concept of "friend-shoring": which is directing supply chains and investment

toward politically aligned partners (Yellen, 2022), was officially introduced as a part of U.S. national security strategy, emphasizing that the main goal of such strategy is to invest into the Science and Technology sector of politically close nations.

The empirical strategy exploits variation in UN General Assembly voting patterns, which serve as a revealed measure of geopolitical alignment. Specifically, the ideal point distance scores developed by Bailey, Strezhnev, and Voeten (2017) offer a systematic and internationally comparable measure of political closeness to the US based on the entire history of UN voting outcomes, not just bilateral diplomacy and alliances. This approach allows for two important things. First, it enables capturing alignment as a continuous measure, thus allowing for differences even among U.S. allies, while also differentiating these from other countries. Second, since UN votes represent revealed policy preferences that apply to various international affairs issues, they are unlikely to be endogenous like bilateral aid or trade flows, which in turn can very well be driven by investment ties (Dreher & Gassebner, 2008; Blonigen & Piger, 2014). The alignment measure is then combined with industry-level foreign direct investment from the U.S., measured using BEA data and World Bank (2024) controls for GDP, openness, and institutional quality.

The findings contribute to three bodies of literature. First, results stretch the field of political economy of foreign direct investment (FDI) in showing how multi-lateral political interactions, beyond bilateral diplomatic relations, influence investment patterns and how the sectoral specificity of these links conforms to a logic of politics rather than economics (Pandya, 2016; Biglaiser & DeRouen, 2006). Second, results contribute to a literature that shows how global geopolitics shapes the fragmentation of global markets in terms of growing political sorting in international trade (Freund et al., 2023) and even private equity investments (Kwon et al., 2024), adding FDI to this list. Third, findings help understand the ongoing policy discussions regarding friend-shoring and preferences for allied nation investments (Sullivan, 2023) by providing an empirical benchmark against which any such policy impact could be measured.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant theoretical and empirical literature. Section 3 describes the data and measurement strategy. Section 4 presents the empirical specification and main results. Section 5 discusses robustness checks and alternative interpretations. Section 6 concludes with policy implications and directions for future research.

2. Literature Review

This paper sits at the intersection of three bodies of literature: the political economy of foreign direct investment, the role of institutions and political risk in shaping capital flows, and the emerging literature on geopolitical fragmentation of global markets. Rather than surveying each

in isolation, this review traces how these strands collectively motivate the present study and identifies the gap it addresses.

2.1 The Political Economy of FDI: Beyond Market Fundamentals

According to the conventional economic theory on FDI, which is based on Dunning's (1981) OLI paradigm and the multinational production model of Helpman (1984), the key drivers of the decision-making process for international investments by enterprises include ownership advantages, locational conditions, such as the size of the market, and low production costs, and incentives for internalization. Both theories make precise forecasts regarding the fact that foreign direct investments will be made in big countries with rapid economic growth and low production costs. Indeed, studies of gravity equations for FDI prove that these variables account for much of its cross-border behavior.

However, despite its robust nature, a growing trend within the literature on FDI is that economic factors account for little of the remaining variance in FDI, especially after accounting for country fixed effects. Such unexplained variance has led to a new emphasis on political economy theories of FDI. Political stability and property rights protection are by now accepted factors influencing the location choice of FDI (Jensen, 2003; Busse & Hefeker, 2007), and there have been numerous findings showing that bilateral political ties, such as alliance formation, common IGO membership, and diplomatic relations, are independent predictors of FDI in their own right, after controlling for economic covariates (Biglaiser & DeRouen, 2006; Gartzke & Quan, 2001). The underlying logic is that firms investing in politically friendly countries minimize the risk of expropriation or other disruptive events, and will benefit from a more receptive attitude towards their investment.

This paper builds directly on this theory but defers from it in an important respect. Most existing work uses bilateral measures of political relationship; treaty membership, alliance portfolios, or aid flows, which are measures that are potentially endogenous to investment. A country that receives large U.S. FDI inflows may be more likely to sign bilateral investment treaties or receive diplomatic attention precisely because of that economic relationship. By contrast, this paper uses UN General Assembly ideal point distances (Bailey et al., 2017) as a multilateral, revealed-preference measure of alignment that is less directly tied to bilateral economic outcomes, offering a cleaner identification strategy.

2.2 Institutions, Political Risk, and Sector Heterogeneity

Secondly, there is literature that posits that the impact of political factors on FDI varies depending on the type of investments being made. In their study, Asiedu and Lien (2011) and Kolstad and Wiig (2012) find out that natural resource FDI is quite insensitive to the quality of institutions – investors will go to wherever the resources can be found even if the country has poor governance practices. This is contrary to the manufacturing and services sector, which is

very sensitive to political and institutional variables. This is because such investments require long-term engagement in the business and integration with host country regulations (Pandya, 2016).

Technology FDI sits at the extreme end of this spectrum. Besides the transfer of physical capital, FDI entails the transmission of knowledge, innovation, and organizational processes that are challenging to safeguard in institutional settings where legal and political constraints prevail. According to Javorcik (2004), Intellectual Property Rights(IPR) protection is a key predictor of technology FDI relative to other industries. Henisz (2000) reaches the same conclusion regarding political risks as defined by the potential for unilateral changes in policies. Specifically, he shows that asset specificity, defined as the level of difficulty involved in relocating resources between industries, plays a decisive role in determining political risks and their impact on foreign investments. As a consequence, the hypothesis suggests that the political risk reduction effect induced by geopolitics will be more significant in technology-intensive industries. The sector-specific approach in this paper thus has theoretical underpinnings, as described above.

The most recent and relevant research examines the geopolitical reorganization of global economic flows. Increasing numbers of scholars highlight the growing role of politics in shaping international economic integration, which had been previously considered to occur according to principles of comparative advantage, through empirical research. Gopinath et al. (2024) show trade fragmentation along geopolitical fault lines due to the increase in geopolitical tensions between the United States and China, with reduced trade volumes among geopolitically distant countries compared to those within geopolitical blocs. Freund et al. (2023) also identify changes in the orientation of supply chains towards politically similar counterparts, especially regarding semiconductors and electronic components. Finally, Kwon et al. (2024), in venture capital research, show the increased sorting of foreign investment by politics since 2016 with decreased investments in China by American firms and redirection of the flows towards allies.

In this regard, the current study seeks to contribute to the new strand of research through an investigation of the role of geopolitics in foreign direct investment in addition to trade flows. Indeed, existing fragmentation studies mostly deal with trade and supply chains; the role of FDI is rarely discussed. In other words, the fact that alignment significantly correlates with technology FDI but not finance or mining FDI implies that fragmentation is taking place in a targeted manner in sectors with high strategic sensitivity, rather than constituting a broader decoupling trend. Thus, when conceptualizing geopolitical fragmentation as an overarching phenomenon, there are specific areas where such trends need to be considered in isolation.

Finally, the present paper offers some insights into policy prescriptions related to friend-shoring and allied-nation investment choices in strategic industries (Yellen, 2022; Sullivan, 2023). Friend-shoring and allied nations' FDI policies suggest that the latter is inherently aligned with

geopolitical interests and needs to target certain countries accordingly. However, the present research represents one of the few attempts at testing if this idea is reflected in the existing reality and whether the decision to invest according to geopolitical logic happens naturally before issuing any official instructions from the authorities.

2.4 Positioning This Paper

Conclusively, the existing literature establishes that political factors matter for FDI, that their effects are sector-specific, and that global investment flows are increasingly organized along geopolitical lines. What remains underexplored is the intersection of these findings: whether geopolitical alignment, measured in a way that is less endogenous than bilateral diplomatic instruments shapes technology FDI specifically, and whether the emerging friend-shoring dynamic is already visible in pre-policy investment data. This paper addresses that gap directly, using a panel identification strategy that allows for comparison across sectors and across fixed-effects specifications to assess the robustness of the alignment effect.

3. Data

This study draws on three primary data sources, merged into an unbalanced panel of 58 countries observed annually from 1999 to 2024. The three sources are the U.S. Bureau of Economic Analysis, the United Nations General Assembly voting database of Bailey, Strezhnev, and Voeten (2017), and the World Bank World Development Indicators (World Bank, 2024). Each is described in turn below.

Several attributes of the data are highlighted in Table 1, each of which holds importance for the statistical analyses to follow. First, the three FDI/GDP series differ substantially in scale and dispersion. The ratio for FDI in science and technology has the lowest mean value, with only 0.313% of the GDP of host countries invested in such projects, due to the nature of U.S. technology investments in the region being highly concentrated among its closest allies. The mean FDI in finance ratio has a value of 3.015%, although it has a high standard deviation of 11.419%, excluding Bermuda; this dispersion occurs because the FDI in finance is disproportionately large relative to GDP in smaller nations where the United States has made more significant financial investments. Mining FDI has an intermediate value of 0.406% with low dispersion. These differences in scale and distributional shape across sectors are themselves informative: if geopolitical alignment were a generic determinant of overall U.S. FDI rather than a sector-specific one, the alignment coefficient should appear uniformly across all three series and this will be talked more about through table 2.

Table 1: Summary Statistics

Variable	N	Mean	SD	Min	Median	Max
Dependent Variables						
FDI: Science & Tech (% of GDP, winsorized)	1,225	0.3132	0.7306	0.0011	0.0892	5.0422
FDI: Finance (% of GDP, winsorized, excl. Bermuda)	1,158	3.0146	11.4185	-0.0396	0.2834	83.502
FDI: Mining (% of GDP, winsorized)	1,060	0.4064	0.7105	-0.0413	0.0424	3.2929
Key Independent Variable						
UN Ideal Point Distance	1,198	2.1806	0.8299	0.0539	2.0554	4.2990
UN Vote Agreement Share	1,198	0.4819	0.1719	0.1042	0.4681	0.9337
Economic Controls						
Log GDP Per Capita	1,225	9.8117	1.0872	6.6101	10.0245	11.7951
Log Trade Openness	1,191	4.3374	0.6076	2.9735	4.2191	6.0807

Notes: Sample covers 58 countries observed annually from 1999 to 2024. FDI variables are U.S. outward foreign direct investment as a percentage of host country GDP, winsorized at the 1st and 99th percentiles. Bermuda is excluded from the Finance series due to its role as a tax haven (FDI/GDP exceeding 1,000% in several years). Negative Tech FDI observations reflecting disinvestment episodes are excluded from the Tech sample. UN Ideal Point Distance is drawn from Bailey, Strezhnev, and Voeten (2017); lower values indicate closer alignment with the United States. Economic controls are drawn from the World Bank World Development Indicators (2024).

3.1 Dependent Variable: U.S. Outward FDI by Industry

The dependent variable in this study is FDI expressed as a percentage of host country GDP rather than in log levels. This scaling removes the mechanical size advantage of larger economies, ensures that the estimated alignment coefficient reflects the economic significance of U.S. technology investment relative to the recipient economy rather than its absolute magnitude, and eliminates the need to log-transform a series that contains negative values reflecting disinvestment episodes. In this paper, three sectors are looked at: Science and Technology, Finance, and Mining. The Science and Technology series is the main dependent variable. The Finance and Mining sectors are chosen as comparison sectors according to the cross-sectoral identification approach introduced in Section 4.

3.2 Key Independent Variable: UN Ideal Point Distance

The primary measure of geopolitical alignment is the ideal point distance (IPD) from each nation to the United States based on Bailey, Strezhnev, and Voeten (2017). The metric is constructed via a dynamic item-response framework using the entire history of UN General Assembly roll-call votes to estimate each member-state's latent ideal point for each session. The difference between the ideal point of a nation and that of the US during each year becomes the measure of geopolitical alignment, where smaller IPDs imply closer alignment. There are a number of benefits to using this variable compared to bilateral diplomatic tools, such as its derivation from revealed voting behavior on all UN resolutions, not being endogenous to economic relations as in the case of treaties, and providing a comparable score to other nations on a common scale.

For the chosen sample, the IPD variable takes a mean of 2.261 and a standard deviation of 0.812 with values ranging from 0.054 to 4.299. As seen in Figure 1, the metric is clearly bimodal with two clusters of nations, namely IPD ≈ 1.5, mainly Western Europe, and IPD ≈ 2.9, Global South/BRICS-aligned.

4. Empirical Strategy

The main empirical challenge of this paper is obvious to state but difficult to resolve: Countries that vote in close alignment with the US in the UN are systematically different from those that do not; they tend to be wealthier, more institutionally stable, more open to trade, and more deeply integrated into the U.S.-led economic and security arrangements. Naive comparisons of FDI flows between countries that are politically aligned with the United States and those that are not will inevitably confound the impact of political alignment with other variables that have been observed to be economically correlated. This paper's empirical strategy attempts to deal with this problem through progressively adding fixed-effects estimators.

4.1 Baseline Specification

The baseline estimating equation takes the following form:

$$\left(\frac{FDI}{GDP}\right)_{it} = \alpha + \beta \cdot IPD_{it} + \gamma X_{it} + \varepsilon_{it}$$

The dependent variable $ln(FDI)_{it}$ is the natural log of the amount of outward Science and Technology FDI by the United States to country i in year t, in million USD. By taking the log transformation, we not only normalize the right-skewed distribution with varying ranges, from low amounts to tens of billions of dollars, but also express the coefficient of interest in proportional terms.

The key Independent Variables is IPD_{it}, which is the UN ideal point distance between country i and the United States in year t. IPD_{it} measures how close country i's position in the international arena is relative to the United States, based on UN General Assembly voting records (Bailey, Strezhnev, & Voeten, 2017). Using the full history of voting behavior of all nations for each UN General Assembly session, an ideal point model is fit with a dynamic voting model to infer the policy positions. The ideal point distance IPD_{it} obtained will be continuous and time-varying, with lower distances indicating greater similarity in voting behavior and hence closeness in interests between both parties.

The vector X_{it} contains a set of standard economic controls drawn from the FDI gravity literature. They consist of the log of GDP, which represents market size; the variable for trade openness as the percentage of exports plus imports to GDP; and an institutional quality index

from the World Bank's Worldwide Governance Indicator, which reflects factors such as the rule of law and quality of regulation. The use of these controls is intended to confirm whether or not the alignment coefficient is simply reflecting the correlation between higher levels of wealth, greater trade openness, and superior institutional governance on the one hand, and closer voting behavior and increased FDI on the other.

4.2 Fixed Effects Specifications

The initial OLS model is useful, although biased due to unobserved country heterogeneity. There exist many attributes among countries such as the country's history with the US, proximity, colonial legacy, and others variables which are not included in X_{it} but are correlated with both alignment and FDI. The preferred specification uses country fixed effects as follows:

$$\left(\frac{FDI}{GDP}\right)_{it} = \alpha_i + \beta \cdot IPD_{it} + \gamma X_{it} + \varepsilon_{it}$$

Where α_i is the fixed intercept unique to each country. It is an extremely stringent specification: the only way β can be estimated, is through within-country variation over time, which is variation between a country's alignment with US votes over time and changes in FDI flows. A country that becomes increasingly aligned with the US over time will attract more technology FDI, whereas one drifting away would get decreasing FDI. Thus, this identification approach is much more convincing as compared to cross-country comparison since it includes all the possible variables that could confound the estimation.

A still more restrictive specification includes year fixed effects too:

$$\left(\frac{FDI}{GDP}\right)_{it} = \alpha_i + \delta_t + \beta \cdot IPD_{it} + \gamma X_{it} + \varepsilon_{it}$$

Where δ_t is a yearly intercept that captures global shocks that are prevalent to all countries in a particular year, such as global investment cycles, the change in monetary policy of the United States, and global FDI from the United States due to domestic reasons. The two-way fixed effects model is the least restrictive and the right one to use when evaluating the robustness of the alignment effect in light of time-varying global shocks.

4.3 Identification Intuition

The logic behind this identification strategy warrants a clear explanation. The question is not whether countries with more alignment receive more FDI in absolute terms because indeed, they do, but whether, after controlling for both a country's fixed effects and the state of international investments in any particular year, an increased degree of alignment leads to increased technology FDI from the United States. This is a much stricter test, and the reduction in the coefficient under two-way fixed effects implies that part of the correlation between alignment

and FDI comes from differences across countries' permanent features or general temporal trends rather than from the effect of alignment within a country.

This attenuation can be considered a valuable result in its own right, as opposed to a weakness of the empirical strategy. It means that while the geopolitical explanation holds true based on pooled estimates and in the presence of only country fixed effects, it does not hold when accounting for all time-varying determinants of investment included in year fixed effects. There are several possible explanations for this result. First, the alignment effect may indeed exist but act through mechanisms that correlate with cycles in global investments, for example, times when there are elevated tensions between the US and China could equally lead to nations aligning with US votes and experiencing investment diversion in favor of allied nations. As such, isolating the alignment effect from its surrounding geopolitical context on the basis of yearly fixed effects might not prove feasible. Secondly, the variation in the alignment of voting patterns in the UN at a country-level could be too small over the study period to yield any statistically significant resulting in attenuation due to insufficient statistical power rather than absence of an effect. Distinguishing between these interpretations is a priority for further work.

4.4 Sector Comparison as Additional Evidence

Another identifying strategy utilizes heterogeneity between industries rather than heterogeneity between countries over time. If the alignment effect is just a proxy for the attractiveness of the country as a destination for U.S. foreign direct investment, then it must be equally prominent in all industries. It is hard to reconcile the large and statistically significant coefficient on IPD in the Science and Technology industry with the very small and statistically insignificant coefficient in Finance and Mining industries under the above alternative theory. Of course, the industry-level test does not help solve the problems outlined above but provides yet another piece of supporting evidence that is consistent with a geopolitically-motivated mechanism specific to technology investment rather than a generic FDI determinant.

4.5 Limitations

There are three notable limitations to the empirical approach that should be highlighted. First, reverse causality continues to be an issue; the large FDI in technology from the United States may itself affect the foreign nation's diplomatic alignment with the United States, thereby creating a reinforcing mechanism whereby larger FDI induces more alignment, hence biasing the OLS coefficient. The inclusion of country fixed effects controls for time-invariant unobservable factors, does not prevent such dynamic changes to occur, and the use of instrumental variables: taking advantage of random variation in UN voting behavior due to rotating memberships or bloc alliances, would offer clearer causal identification of the relationship investigated here. Second, because of confidentiality concerns, some observations of the FDI variable are suppressed when there are few reporting companies in particular country-year pairs. Third, and most importantly, UN ideal point scores measure ideal points for all UN resolutions rather than those pertaining to

technologies or economic policies; thus a country's ideal points in general may not accurately represent its relationship with the United States with respect to technology.

5. Results

5.1 Baseline OLS Results

Table 2: Regression Results — Dependent Variable: U.S. FDI as % of Host Country GDP

	Science & Tech FDI/GDP			Finance FDI/GDP	Mining FDI/GDP	Science & Tech FDI/GDP
	(1) OLS	(2) Country FE	(3) Two-Way FE	(4) Country FE	(5) Country FE	(6) Post-2008
Key Variable						
Ideal Point Distance	0.122***	−0.150**	−0.184**	−0.406	−0.205	−0.060
	(0.046)	(0.072)	(0.089)	(0.812)	(0.169)	(0.109)
Controls						
Log GDP Per Capita	0.211***	0.051	−0.379	0.881*	−0.141	−0.191
	(0.040)	(0.097)	(0.292)	(0.513)	(0.120)	(0.193)
Log Trade Openness	0.277***	0.266	0.099	2.080	0.316**	0.218
	(0.052)	(0.180)	(0.216)	(1.824)	(0.141)	(0.227)
Post-2008						0.125
						(0.190)
IPD × Post-2008						0.022
						(0.093)
Constant	−3.222***					
	(0.503)					
Fixed Effects						
Country FE	No	Yes	Yes	Yes	Yes	Yes
Year FE	No	No	Yes	No	No	No
Fit Statistics						
Observations	1,164	1,164	1,164	1,101	1,006	1,164
R^2	0.172	0.647	0.666	0.708	0.775	0.654
Within R^2		0.018	0.020	0.008	0.038	0.039

Notes: Standard errors in parentheses. Columns (2)–(6) use country-clustered standard errors. Column (1) uses heteroskedasticity-robust standard errors. The dependent variable is sector FDI as a percentage of host country GDP, winsorized at the 1st and 99th percentiles. Bermuda is excluded from the Finance specification (column 4). Negative Tech FDI observations (disinvestment episodes) are excluded from columns (1)–(3) and (6). Column (1) OLS estimate is subject to upward bias due to correlation between ideal point distance and GDP per capita ($r = -0.54$). Preferred specification is column (2). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The coefficients from six different specifications of the regression are presented in Table 2. Under the pooled OLS baseline specification, the coefficient for the independent variable "UN Ideal Point Distance" is estimated to be +0.122 (SE = 0.046, p-value < 0.01). This implies that countries farther away from the United States in terms of the UN voting scale would receive a greater amount of Science and Technology Foreign Direct Investment (FDI), as a proportion of their GDP. This finding goes against the predictions made under the geopolitical perspective. However, this result cannot be substantiated because the ideal point distance is correlated with per capita income (r = -0.54). Since ideal point distance negatively correlates with the dependent variable GDP, it leads to a bias in the estimate where poorer nations, who are farther from the United States, will have low values of GDP and hence higher values of FDI/GDP ratio, pushing the OLS coefficient in the positive direction even when the true within-country relationship runs the other way.

To put the magnitude of the bias in perspective: moving from the sample mean IPD of 2.181 to the 75th percentile of approximately 2.861, a shift of 0.68 units, the OLS estimate predicts an

increase of approximately 0.083 percentage points of GDP in Science and Technology FDI received. This positive prediction is the artifact of the income-alignment correlation rather than a genuine geopolitical effect, and is reversed once country fixed effects are introduced in the preferred specification.

5.2 Fixed Effects

Controlling for country fixed effects, accounting for all time-invariant differences across countries, results in a coefficient of ideal point distance of −0.150 (standard error [SE] = 0.072, $p < 0.05$), reversing the sign of the OLS estimator. This is the main result of the paper. In the OLS model, the positive coefficient was driven by the mechanical bias created by the income-alignment correlation. Countries that were poorer and more misaligned had smaller GDPs, resulting in higher FDI/GDP ratios in the cross-section, driving the coefficient in the wrong direction. When we account for all time-invariant differences between countries, including the differences in income levels, quality of institutions, historical relationships with the United States, and geography through the inclusion of country fixed effects, the real effect becomes negative. At this point, identification is driven by time-variation in the data for each individual country. The coefficient identifies how changes in alignment toward the U.S. votes affect the ratio of U.S. Science and Technology FDI to GDP in a given country. In economic terms, a one-unit decrease in ideal point distance is associated with an increase of 0.150 percentage points of GDP in Science and Technology FDI received, approximately a 48% increase relative to the sample mean of 0.313%.

The strictest specification incorporates year-fixed effects along with the country-fixed effects, controlling for time-invariant country effects as well as the global shocks common to all countries during each particular year. As indicated in the third column of Table 2, the coefficient on IPD is estimated to be -0.184 (SE = 0.089, $p < 0.05$), which is relatively larger in size compared to that obtained with the country FE model and statistically significant. This finding sharply contrasts with the results based on the log-level specification in earlier versions of this paper, in which case the estimate from the two-way fixed effects specification converged to zero and became insignificant. The discrepancy can be attributed to the GDP normalization of the dependent variable. The coefficient implies that a one-unit decrease in ideal point distance is associated with an increase of 0.184 percentage points of GDP in Science and Technology FDI once both country and year fixed effects are controlled for, a result that strengthens rather than weakens the geopolitical hypothesis relative to the country FE specification. Figure 2 plots the IPD coefficient and its 95% confidence interval across all three Tech specifications, making the progression from the biased positive OLS estimate through the negative and significant fixed effects results

5.3 Sector Comparison

One important prediction of the geopolitical theory is that the alignment effect would be more pronounced in strategically important industries as opposed to being pervasive in the FDI activity in general. Comparing Finance and Mining FDIs with country fixed effects, both have coefficients of the ideal point distance of −0.406 (SE = 0.812, $p = 0.617$) and −0.205 (SE = 0.169, $p = 0.225$), respectively, neither of which is significantly different from zero, while the significant Tech coefficient was −0.150 (SE = 0.072, $p < 0.05$). Such sector-specific results help dispel the main competing hypothesis. That is, if the alignment effect captured some kind of overall national attractiveness correlated with the UN votes, then it would show up with a positive sign and similar significance level in each of the industries studied. The fact that it is concentrated precisely in the sector where geopolitical considerations are most theoretically salient and absent in Finance and Mining, is consistent with a strategic rather than purely economic investment logic, a contrast that Figure 4 makes visually explicit by plotting the IPD coefficient and its 95% confidence interval across all three sectors side by side under equivalent specifications.

5.4 Post-2008 Interaction

To test whether the influence of alignment on FDI became stronger after the global financial crisis – a period that saw rising strategic competition between the United States and China and increasing focus on technological security in policymaking – a Post-2008 interaction term is included in Model 6 of Table 2. This interaction term has a coefficient of +0.022 (SE = 0.093, $p = 0.813$). It is insignificant, small in size, and positive, whereas the coefficient of IPD without the interaction drops to −0.060 (SE = 0.109, $p = 0.582$). The null findings run contrary to the expectation and must be qualified on account of two limitations. Firstly, the Post-2008 dummy incorporates a significant portion of intra-country variation within the framework of country fixed effects, and not much variation remains for the identification of the heterogeneous impact of alignment on FDI before and after the 2008 crisis. Second, as illustrated in Figure 3b, the largest change in mean ideal point distance takes place after 2021 instead of 2008, implying that if there is a structural break in the association between alignment and foreign direct investment, it might be found closer to the time of heightened technology rivalry between the United States and China and the enactment of FIRRMA in 2018.

6. Discussions and Limitations

The findings of this paper show a strong relationship between UN voting alignment and US FDI into science and technology in developing economies. However, there are three important caveats in relation to the causal relationships that could be inferred from the data.

The first one refers to the reversal of signs in the OLS model shown in Section 5. The GDP-normalized OLS regression produces a positive and significant coefficient on ideal point distance, contrary to the geopolitical model. However, this finding should be understood not in

its substantive sense, but rather in its diagnostic value. It simply illustrates the very strong negative relationship between GDP per capita and ideal point distance (r = -0.54). Poorer and more divergent countries will thus tend to look like they are receiving higher shares of FDI compared to their GDP in the cross-sectional comparison. This example shows that the problem of confounding cannot be fully addressed through scaling FDI by GDP without using country-fixed effects, while any cross-sectional comparison of country FDI shares remains biased. Country fixed-effects are required to identify the real alignment effect within each country. The consistency of the negative and significant coefficient across both the country FE and two-way FE models in the GDP-normalized regression, which is clearly much more credible than the log-level estimation, constitutes a significantly stronger identification strategy than what was possible in the log-level regression, where the two-way FE estimate became insignificant.

The second limitation is reverse causality. States that receive high levels of U.S. technology investment due to increased economic integration could develop a greater affinity towards U.S. policy through their diplomatic votes. Dealing with this issue requires using an instrumental variable strategy that involves using an exogenous variation in UN votes through, for example, rotating Security Council members. Unfortunately, this is not within the scope of the current study; however, it should be done as the natural follow-up on the present research.

Thirdly, there is the issue of non-randomly missing data in the BEA database. Small cells containing few respondents are automatically omitted due to the need to maintain confidentiality. These happen to be the same countries that would get relatively little foreign direct investment due to their low geopolitical affinity with the United States. Therefore, the sample used can be biased toward already having considerable amounts of foreign direct investment in the United States. Nevertheless, the sector specificity of results is a powerful indicator of their validity. The alignment effect for science and technology is large and statistically significant, while it is insignificant for finance and mining sectors.

The honest characterization of these findings is therefore one of a strong, sector-specific association that survives within-country identification but requires stronger causal machinery to interpret as a structural effect of geopolitical alignment on investment.

7. Conclusion

This paper set out to answer a straightforward but underexplored question: does geopolitical alignment with the United States, as revealed by UN General Assembly voting behavior, influence the destination of U.S. outward foreign direct investment in the Science and Technology sector? Using a panel of 58 countries observed annually from 1999 to 2024, and exploiting variation in UN ideal point distances developed by Bailey, Strezhnev, and Voeten (2017).

All three regression models, the OLS, country fixed effects model, and two-way fixed effects model, are telling a coherent story in economic terms once the corrections have been made for the GDP scaling factor and the OLS bias. Once adjusted for the upward bias in the OLS regression due to the negative correlation between ideal point distance and GDP per capita ($r = -0.54$) resulting in a spurious coefficient, it is shown that for every additional unit of ideal point distance, there is a reduction in the amount of U.S. Science and Technology FDI in the host countries is measured as a percent of their GDP by a margin of 0.150 to 0.184 percentage points, which is statistically significant in both the country fixed effects and two-way fixed effects models. The finding continues to hold with the addition of year fixed effects, a more robust specification than had been possible with previous log-linear estimates, and there is no parallel finding for the relationship between finance and mining investments, suggesting that the alignment-investment nexus is specific to the industry where geopolitical concerns would have the greatest theoretical relevance. The results speak to an emerging literature examining how geopolitics is altering global economic relationships (Gopinath et al., 2024; Freund et al., 2023) and add foreign direct investment to the list of economically-relevant avenues through which political alignment influences economic activity.

However, the results require interpretation with caution. The first concern stems from the OLS coefficient sign reversal, which shows that even GDP-weighted FDI measures are vulnerable to cross-section endogeneity, emphasizing that within-country fixed effects should be treated as an essential step in the empirical analysis. Second, the issue of reverse causality still remains relevant; countries that receive more technology investments from the United States may align their foreign policies more with the interests of the superpower due to stronger economic ties, and an instrumental variable approach is required to confirm the direction of causality. Finally, the data suppression policy of the BEA introduces non-random missingness, likely excluding small countries with lower alignment scores from the analysis and thus creating an overestimated effect on alignment among the remaining countries. While this study does not ignore the limitations, it does not exaggerate its contribution either; it simply states explicitly what conclusions can be drawn from the evidence: a significant and robust relationship between geopolitical alignment and U.S. technology investments within sectors in certain countries, which supports the geopolitical explanation.

Nevertheless, its implications are significant. When political alignment, whether partial or indirect, influences the choice of locations where American companies develop technological capabilities abroad, the geographical allocation of innovations, productive abilities, and economic growth beyond comparative advantages is no longer determined by it. In a world where American policy seeks to redirect technology investments to allied nations through friend-shoring strategies (Yellen, 2022) and foreign investment reviews (FIRRMA, 2018), determining the initial effect of political alignment on technology FDIs is necessary to evaluate whether such policy measures are redundant, supportive, or revolutionary.

There are three possible avenues for future research that arise from this study. The first possibility would be the use of an instrumental variables approach by utilizing rotating membership on the UN Security Council or regional voting blocs as instruments for bilateral alignment. This would provide the causal identification mechanism currently lacking from the empirical strategy used in this paper. The second possibility would be to extend the analysis to examine how investment decisions of individual firms behave in order to determine whether there is a difference between effects on the intensive margin (less money invested by individual firms) and extensive margin (fewer firms investing in countries that do not align with their own). Finally, the third avenue to explore would be whether the friend-shoring-foreign direct investment link has increased since 2021 because this period marks the start of friend-shoring policy discussions.

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Appendix

Figure 1: Distribution of UN Voting Alignment

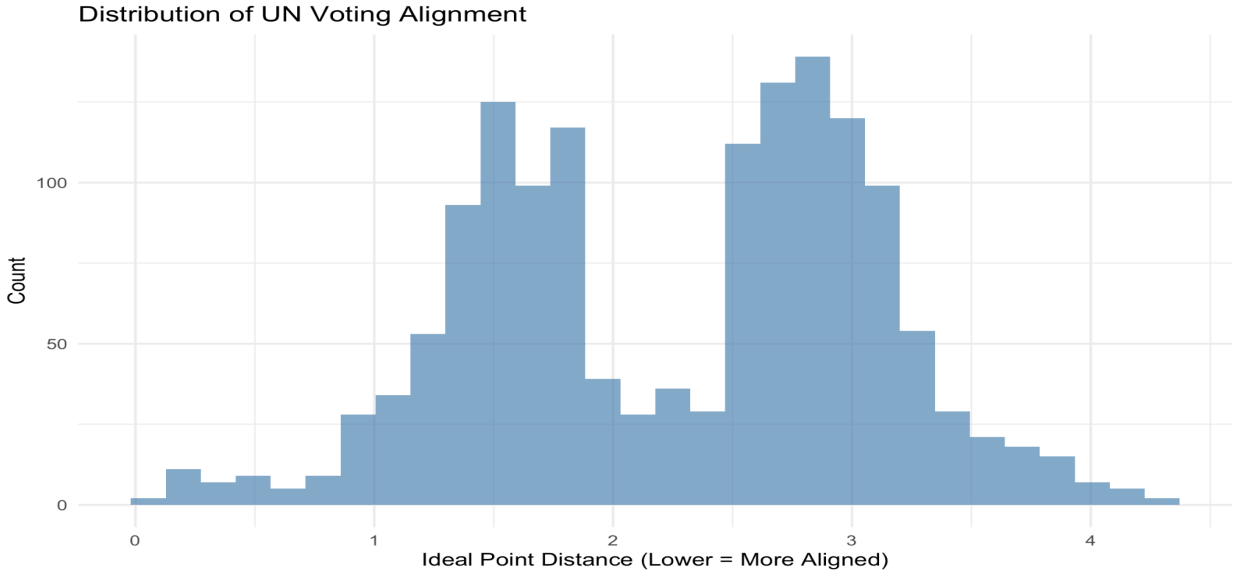


Figure 2:

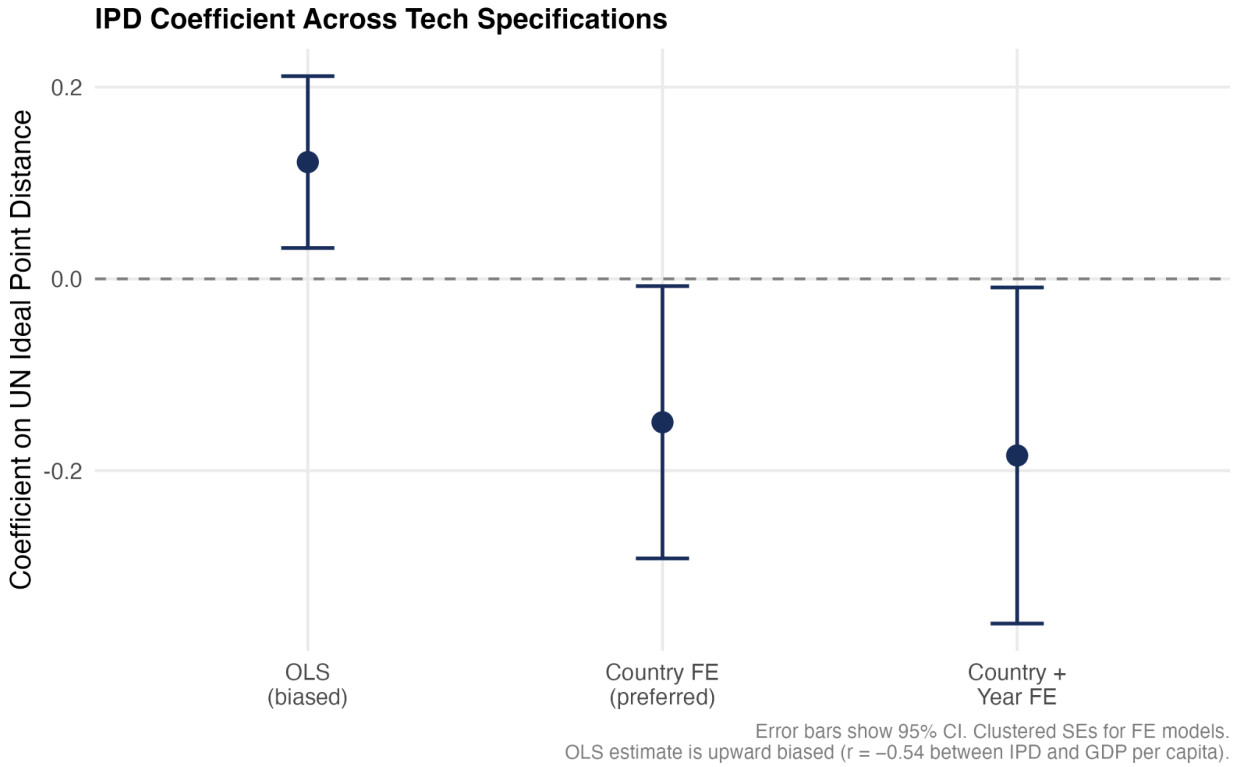


Figure 3a:

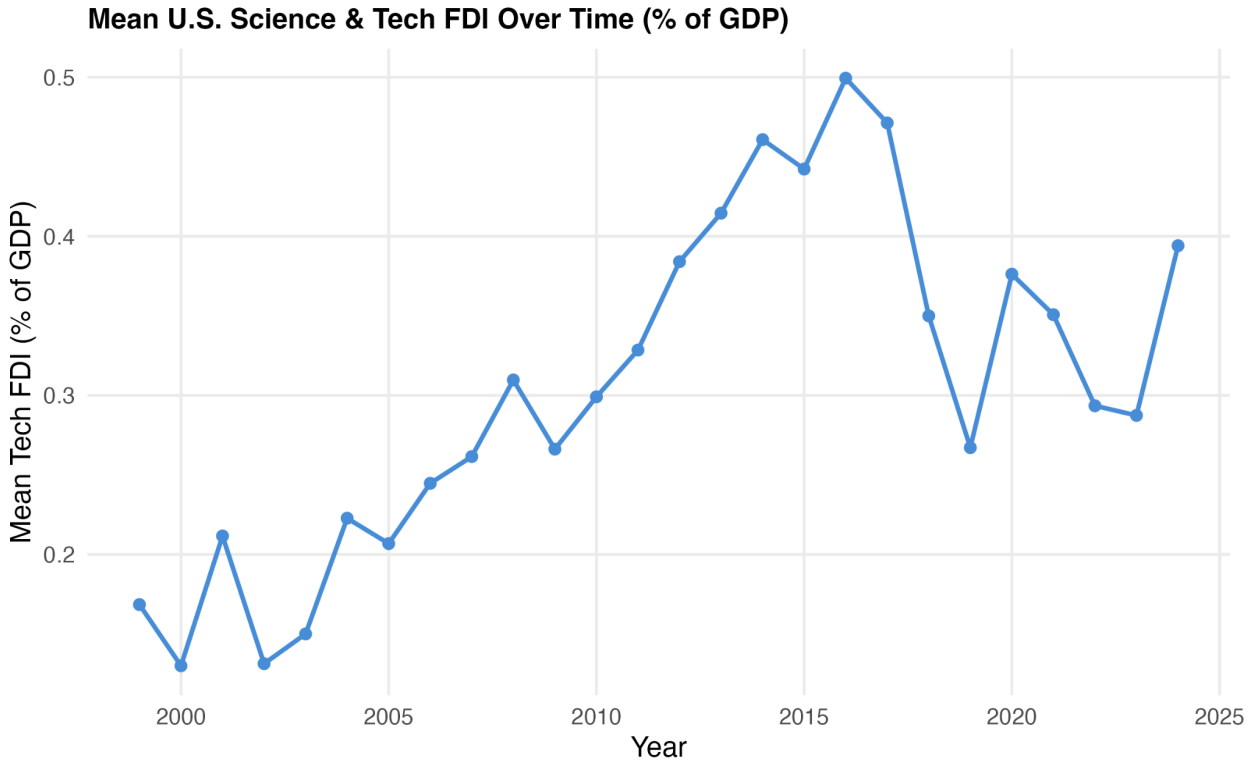


Figure 3b: Ideal Point Distance Overtime

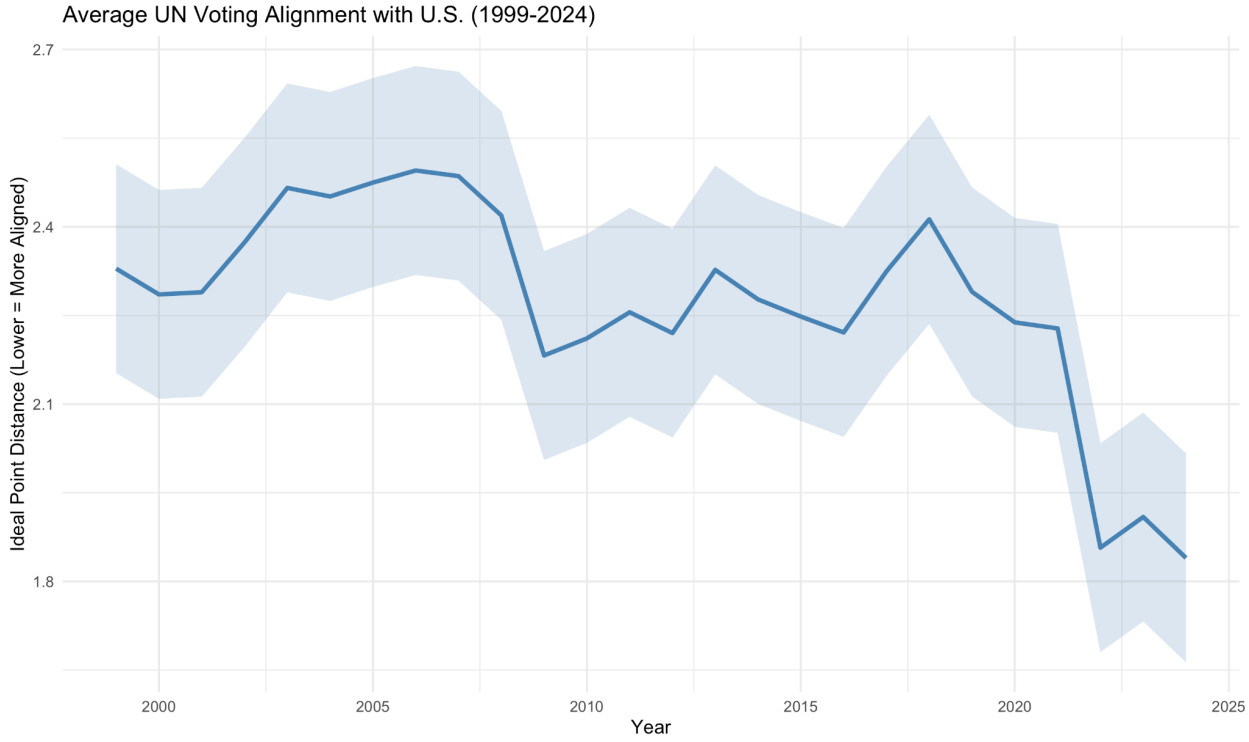
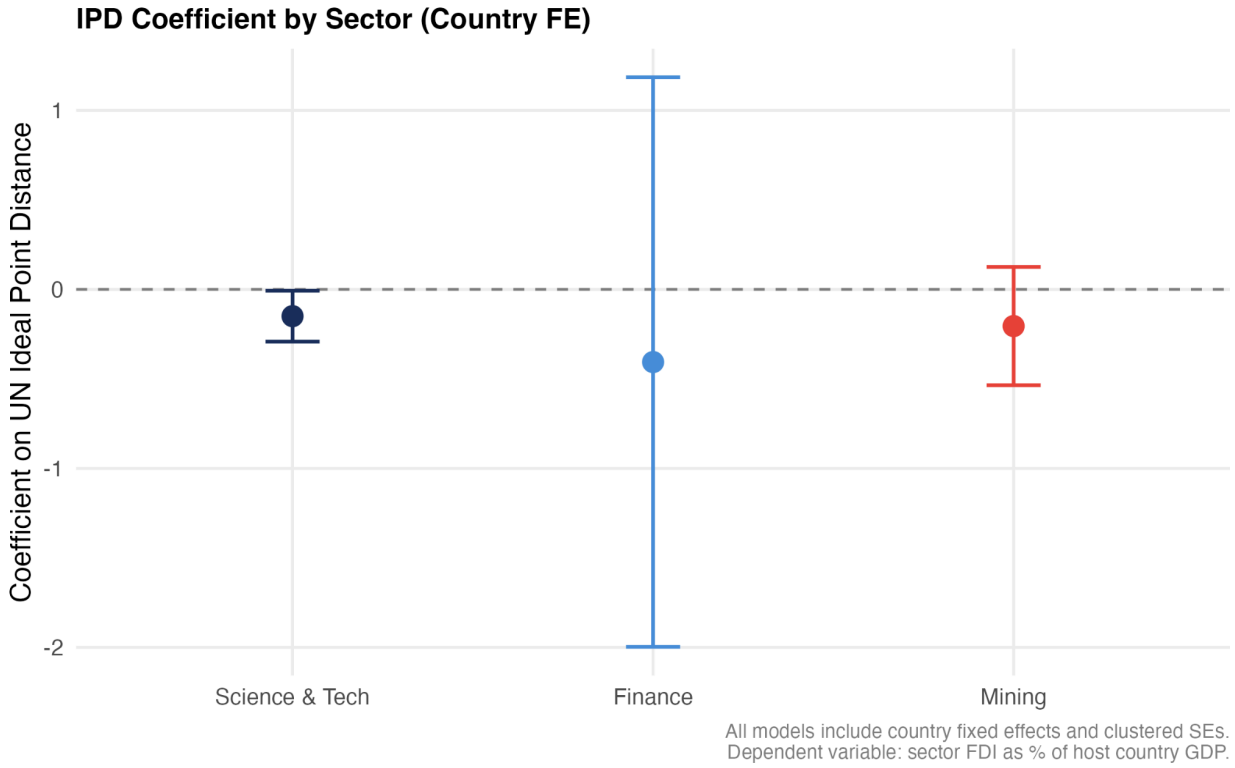


Figure 4:


IPD Coefficient by Sector (Country FE)
Coefficient on UN Ideal Point Distance
1
0
-1
-2
Science & Tech
Finance
Mining
All models include country fixed effects and clustered SEs.
Dependent variable: sector FDI as % of host country GDP.